Brian Pulfer

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Work Authorization: U.S. O-1 visa

Experience

Research Scientist, Instagram Reels Translation — Oct. 2026 – Present
Meta Platforms, Inc. — *Menlo Park, California*

Research Scientist Intern — June 2025 – September 2025
Meta Platforms, Inc. — *Menlo Park, California*

- NVFP4 quantization of LLMs on the vLLM inference engine, cutting memory by **67.5%**/**37.6%** (vs. BF16/FP8) with $< 2\%$ accuracy drop.
- Exploratory studies on NVFP4 quantization of KV-cache in vLLM.

Graduate Researcher (Ph.D. in Machine Learning) — November 2021 – September 2026
University of Geneva — *Geneva, Switzerland*

- Dissertation on self-supervised vision models and robustness to adversarial attacks; published at CVPRW 2025 and ECCV 2024.
- Teaching Assistant for Algorithms and Data Structures courses.

Education

UNIGE - University of Geneva — Geneva, Switzerland
Ph.D. in Machine Learning — *Nov. 2021 – Sep. 2026*

USI - University of Southern Switzerland — Lugano, Switzerland
MSc in Artificial Intelligence (GPA 9.1/10) — *Aug. 2019 – Jun 2021*

SUPSI - University of Applied Sciences of Southern Switzerland — Manno, Switzerland
BSc in Computer Science — *Aug. 2016 – Jun 2019*

Selected Publications [Full list: Google Scholar]

- **B. Pulfer** et al. *Task-Agnostic Attacks Against Vision Foundation Models.* CVPRW 2025 (AdvML Workshop).
- **B. Pulfer** et al. *Robustness Tokens: Towards Adversarial Robustness of Transformers.* ECCV 2024.
- A. Stocco, **B. Pulfer**, P. Tonella. *Mind the Gap! Transferability of Virtual vs. Physical-world Testing of Autonomous Driving Systems.* IEEE TSE.

Honors & Service

Reviewer for CVPR 2025; Judge at the Neural Wave hackathon; SODESKA merit scholarship (USI, 2021); 3rd place, NeurIPS Weather4cast Challenge 2022; Winner, HackZürich 2023.

Technical Skills

Languages: Python, C++/CUDA
Frameworks: PyTorch, vLLM, Hugging Face
Areas: LLM inference & quantization, machine translation, adversarial robustness, distributed training